Christof Teuscher

ECE 410/510: Hardware for AI and ML, Spring 2026

Codefest #1

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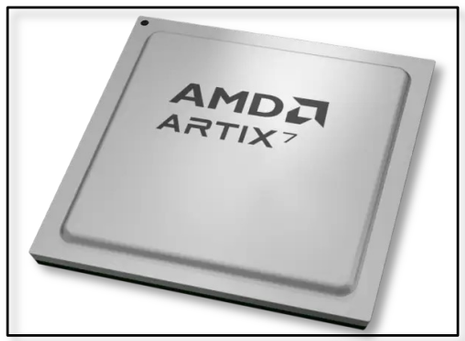

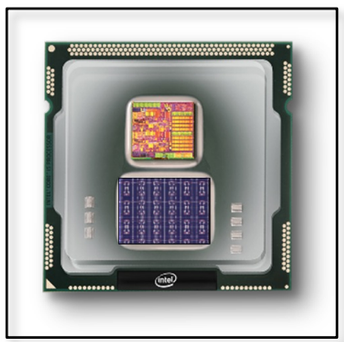

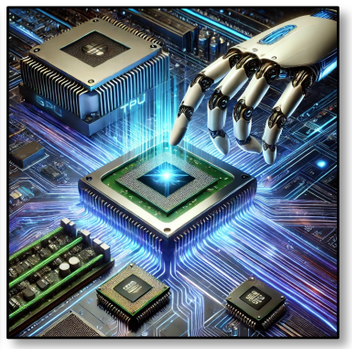

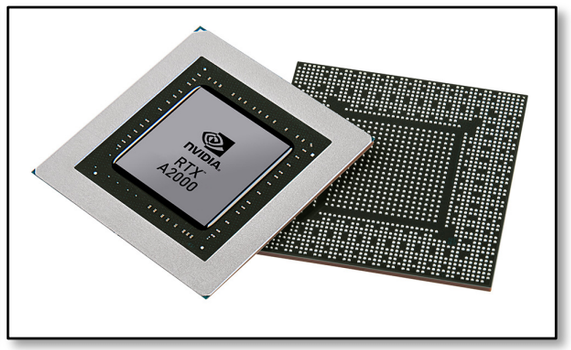

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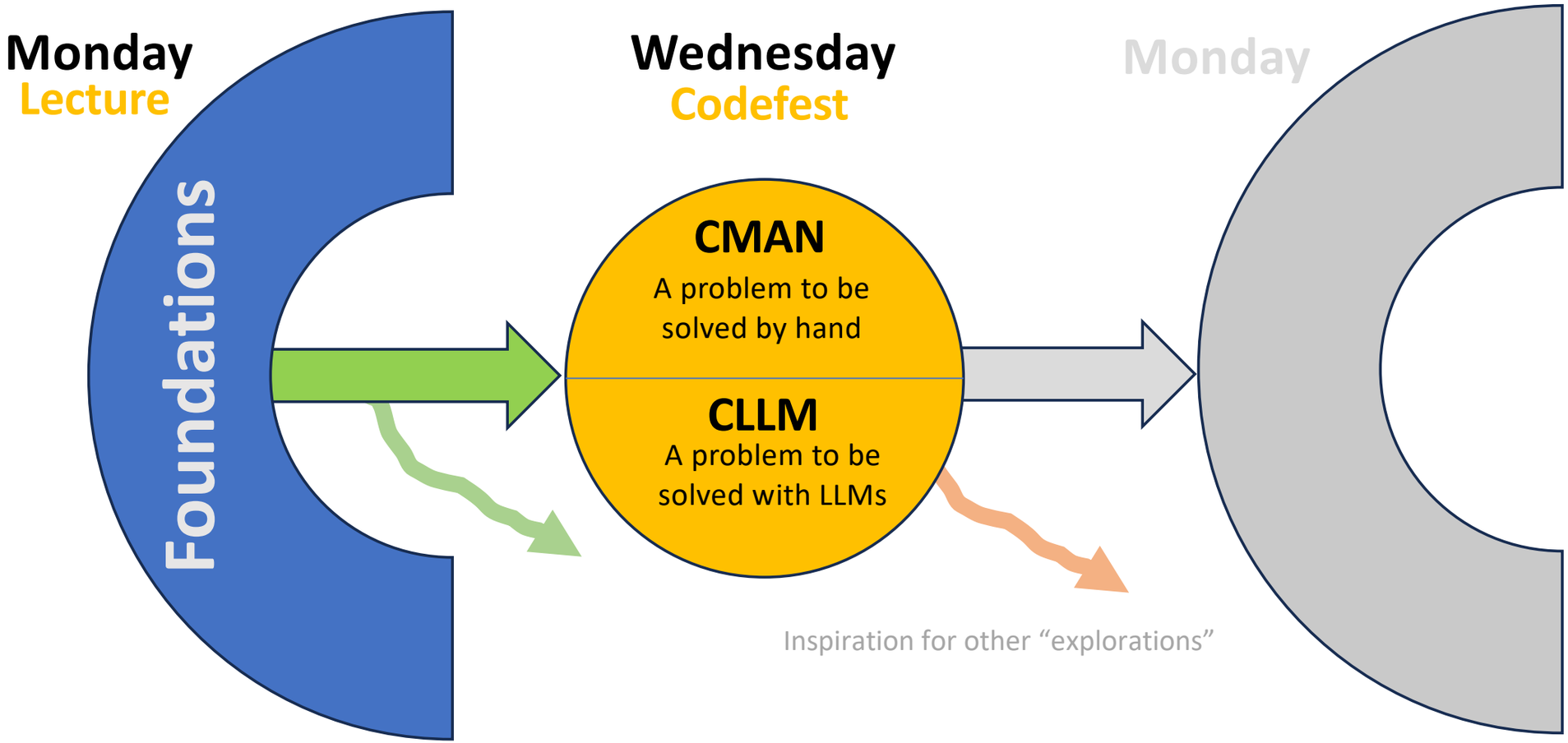
Codefest goals

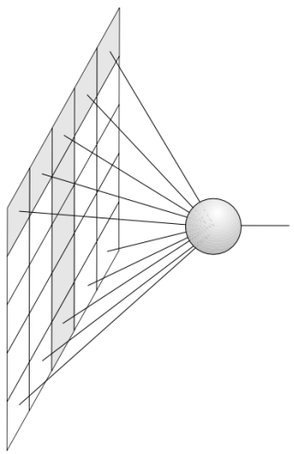
- **Rapid prototyping**, solve problems, and create complex designs using “vibe coding.”
- Learn how to use **LLMs** efficiently to create complex design we couldn’t create without.
- Use LLMs to learn and to gain a deeper understanding of the subject matter.
- **The challenges are supposed to be supporting building blocks for the main project.**
- Collaborate, share and learn from each other.
- Use Slack to ask questions if you are stuck.
- Present, document, communicate.
- **Motivate you to go home and brings things to the next level.**

A typical week



An abstract neuron

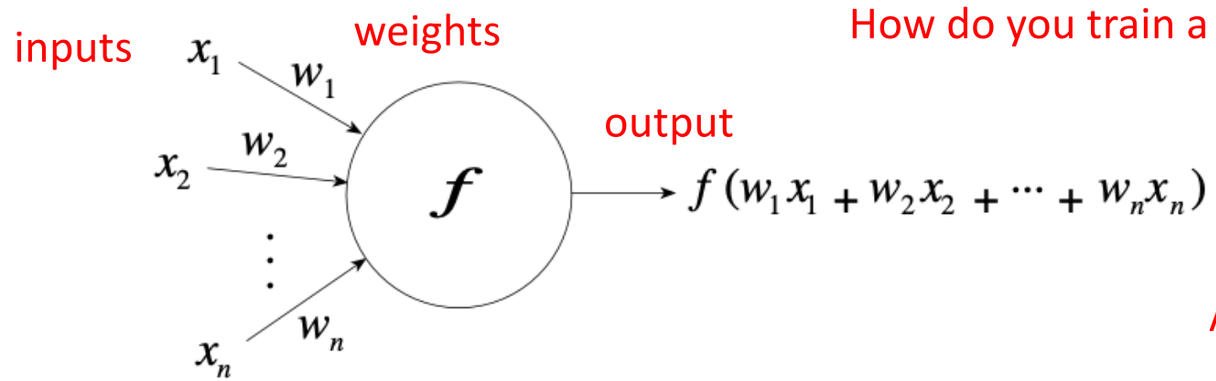
How do you train a neural network?



pattern

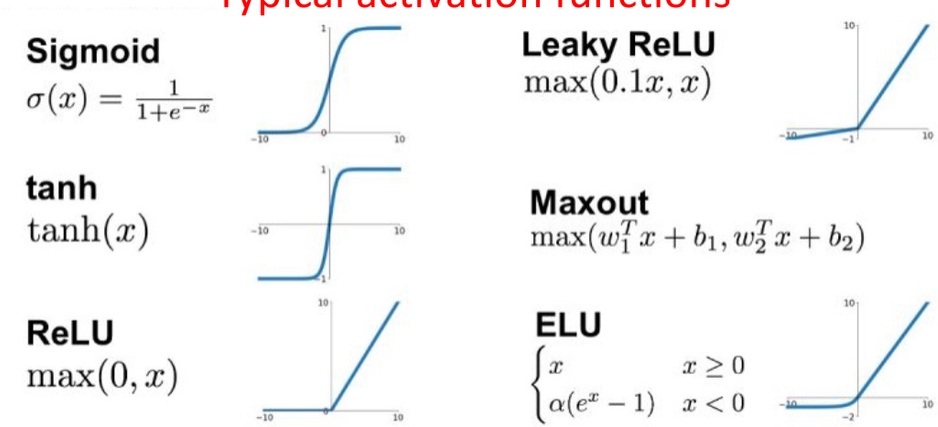
weights

1	1	1	1	1
–1	–1	1	–1	–1
–1	–1	1	–1	–1
–1	–1	1	–1	–1
–1	–1	1	–1	–1



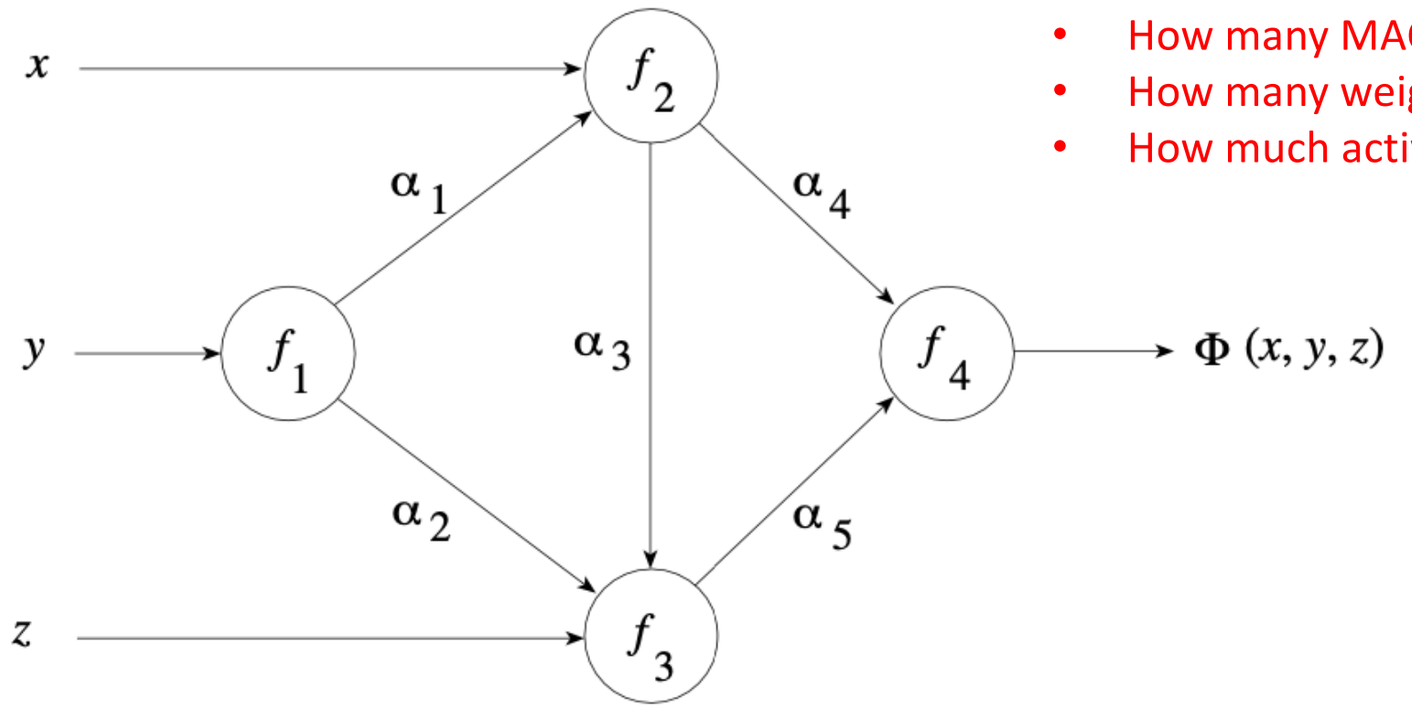
Activation function

Typical activation functions



Rojas, 1986

A neural network



- How many MACs?
- How many weights?
- How much activation memory?

Rojas, 1986

Special types of neural networks

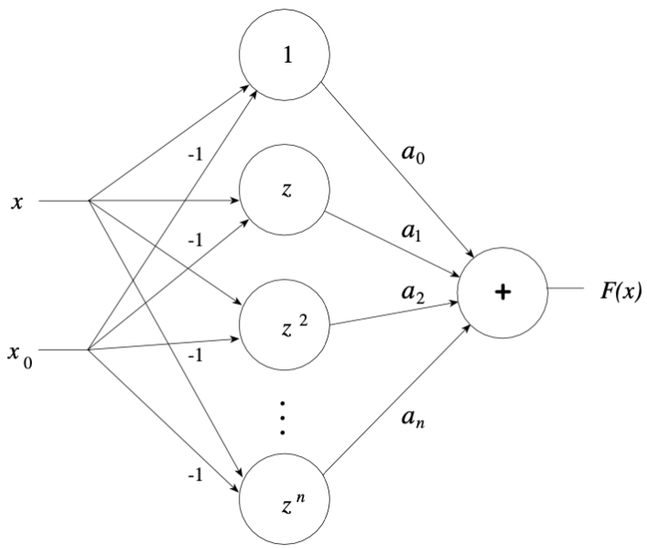


Fig. 1.16. A Taylor network

$$F(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_n(x - x_0)^n + \cdots,$$

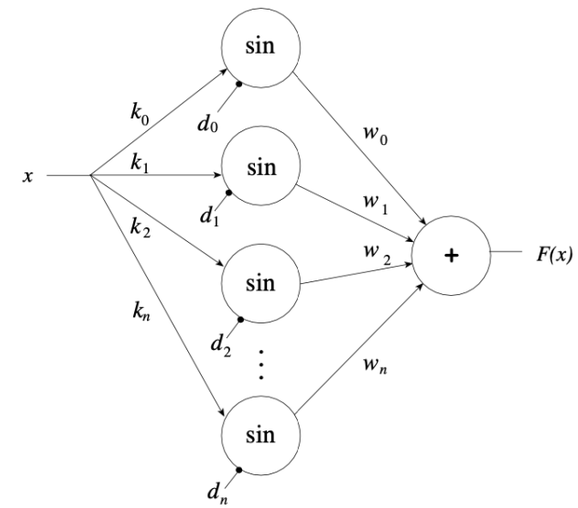


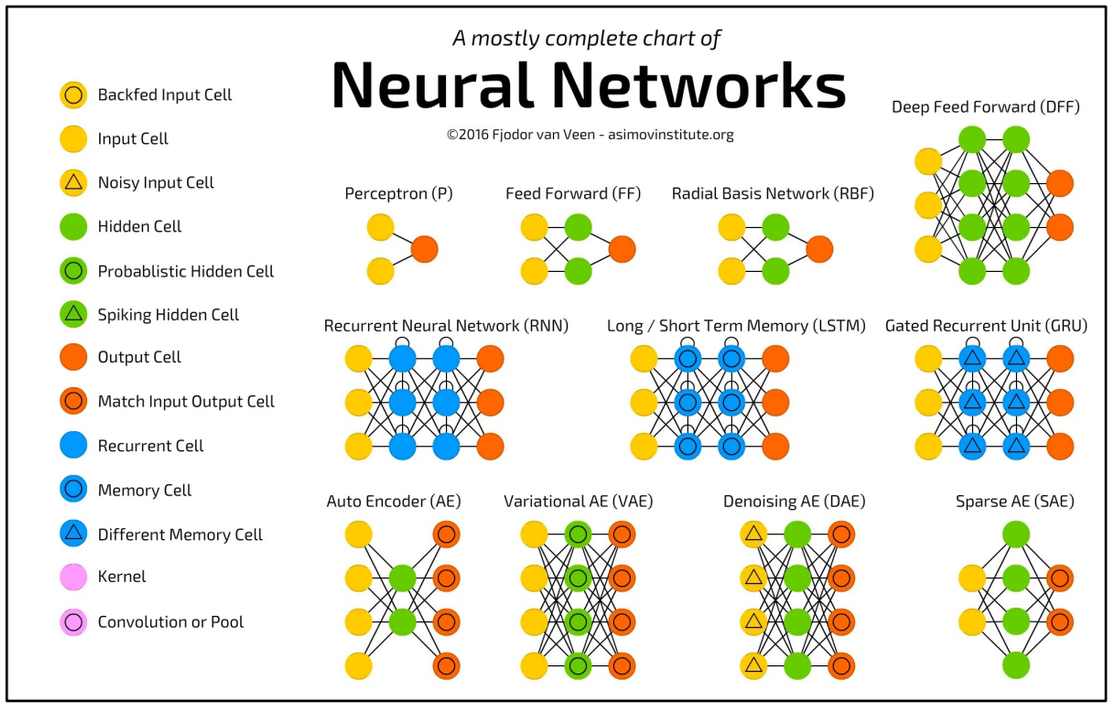
Fig. 1.17. A Fourier network

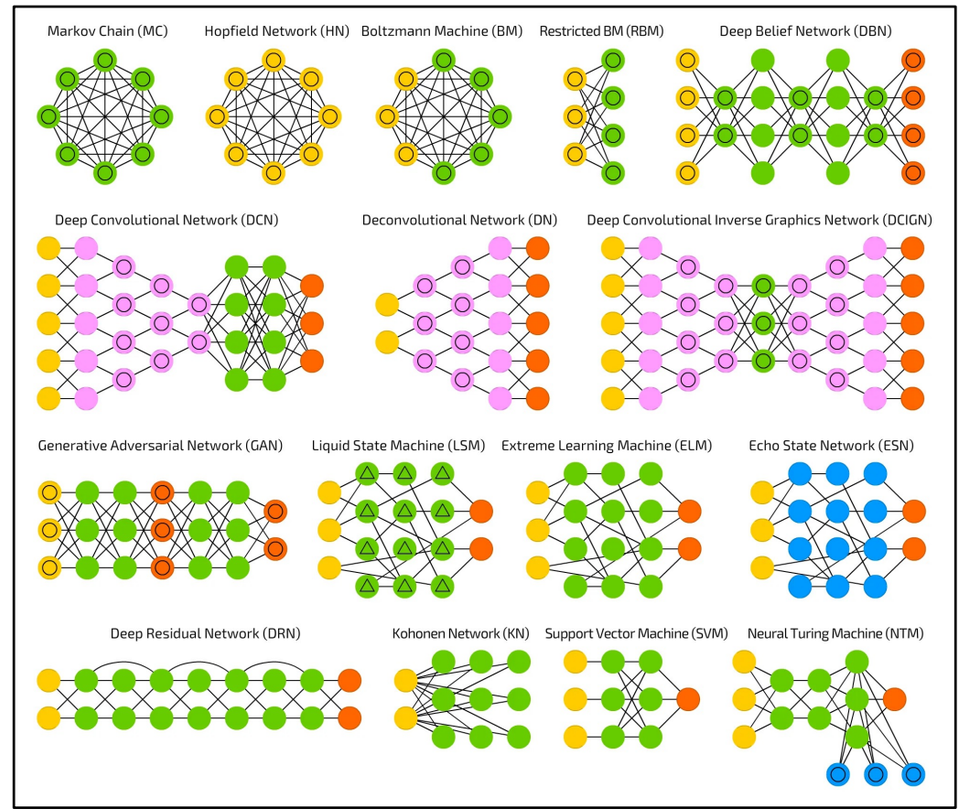
Figure 1.17 shows how a Fourier series can be implemented as a neural network. If the function F is to be developed as a Fourier series it has the form

$$F(x) = \sum_{i=0}^{\infty}(a_i \cos(ix) + b_i \sin(ix)). \tag{1.2}$$

Rojas, 1986

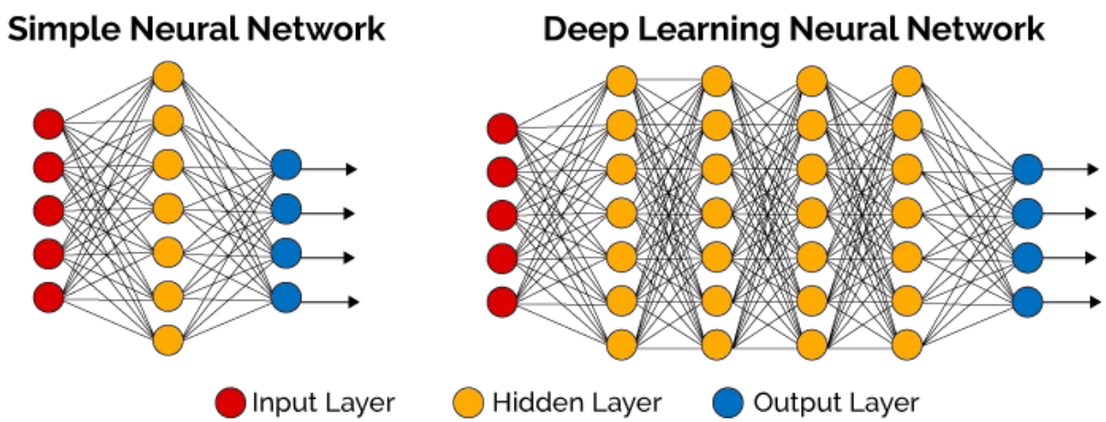
Types of neural networks

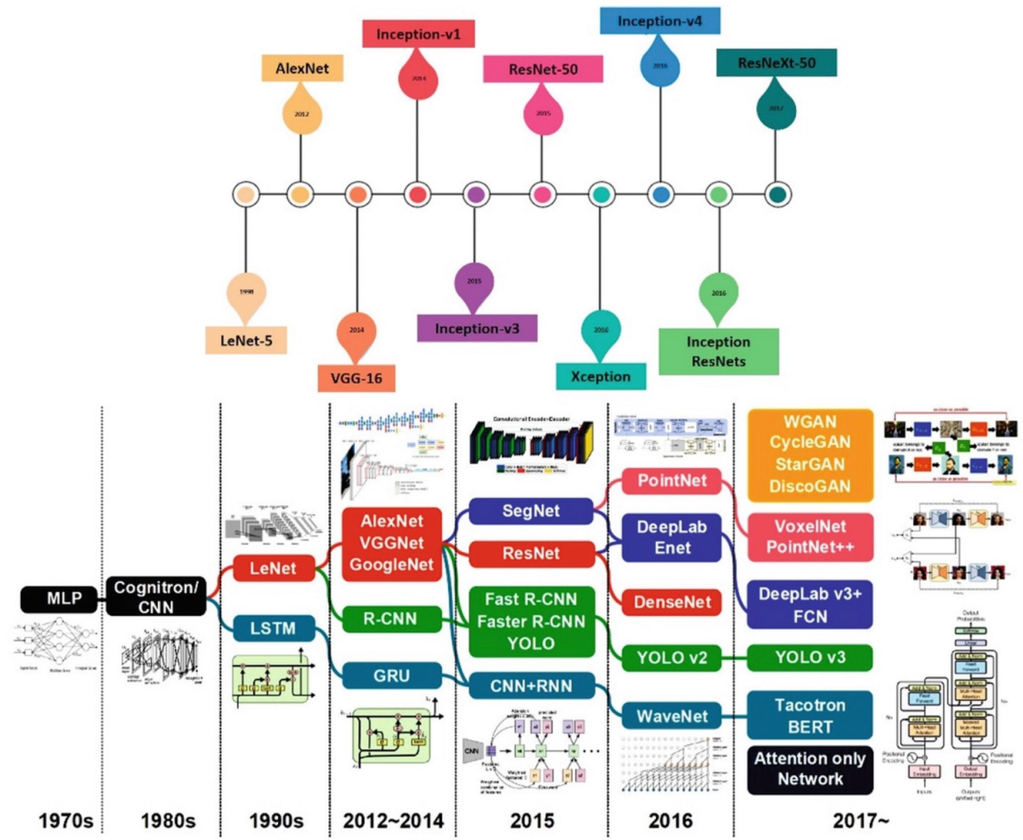




(Fjodor van Veen, 2016, asimovinstitute.org)

Deep neural networks





Today (1)

1. If you haven't yet, set up your Github. That is where all your deliverables need to sit.
2. Make it public.
3. **Goal**: Create an impressive portfolio of challenges you solved in this class, including the final project, that you can showcase to employers.

Today (2)

1. Complete the first challenge (CMAN) by hand.
2. Complete the second challenge (CLLM) with any tool(s) you want to use.
3. Start with the project:
   1. Pick an AI algorithm. Post the algorithm choice in the spreadsheet linked on Canvas.
   2. Answer the first 3 Heilmeier questions.
   3. Draw a high-level diagram of your target algorithm (use tools!).
4. Use Slack to ask questions. Collaborate, help each other…
5. Push all solutions to your Github (no later than Sun, 11:59). Respect the folder/file naming conventions!